Chemistry GCE A Level 2025 - Paper 2

Duration: 3h | Cameroon GCE Board

Time allowed: three hours. Answer ALL the SIX questions in this booklet. The mark allocation is indicated for each question. Each question carries 20 marks. Verify that this booklet contains six questions, no questions are repeated and there are no blank pages. Inform the invigilator in case this booklet contains less than six questions, questions are repeated or there are blank pages so that the booklet should be changed. Blank spaces in this question booklet may be used for rough work. In calculations you are advised to show all the steps in your working, giving your answer at each stage. All necessary working must be shown. No marks will be awarded for answers without brief statements showing how the answers have been obtained. Calculators may be used.

Useful data: relative atomic masses (RAM): C = 12.0, O = 16.0, H = 1.0, Na = 23.0.

The paper has three sections: Section A - Physical and General Chemistry (questions 1 and 2); Section B - Inorganic Chemistry (questions 3 and 4); Section C - Organic Chemistry (questions 5 and 6).

Exercise 1 (20 marks)

a. What is a redox reaction? *[1 mark]*

b. $31.0\ \text{cm}^3$ of an acidified KMnO_4 solution was required to completely oxidize $50\ \text{cm}^3$ of $0.25\ \text{M}$ solution of oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$). The unbalanced ionic equation is:

$$\text{C}_2\text{O}_4^{2-}(\text{aq}) + \text{MnO}_4^-(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{Mn}^{2+}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$$

(i) Write down the half equation for

A: Oxidation:

B: Reduction:

(ii) Write down the overall balanced equation

(iii) Calculate the concentration of the KMnO_4 solution. *[5 marks]*

c. Bond energy can provide information about the energy changes that accompany a chemical reaction.

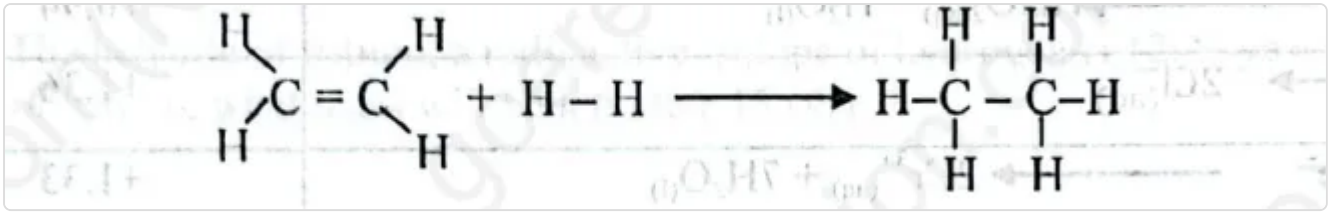
(i) What do you understand by the term bond energy?

(ii) The table below shows some average bond enthalpies

Bond	**Average bond enthalpy/kJ mol^{-1}**
C - C	350
C = C	610

Bond	Average bond enthalpy/$\mathrm{kJ\ mol^{-1}}$
H - H	436
C - H	410

A: Use this information to calculate the enthalpy change for the process



(ethene plus hydrogen giving ethane: $\mathrm{CH_2{=}CH_2 + H{-}H \rightarrow CH_3{-}CH_3}$)

B: The enthalpy change of this reaction was found by experiment to be $-136\ \mathrm{kJ\ mol^{-1}}$. Explain why this value is different from that calculated above. *[4 marks]*

d. The solid state of matter is characterized by the arrangement in fixed patterns of its constituent particles (atoms, ions, or molecules). Complete the table below by filling in either the crystal type or example.

Crystal type	Example
Body centred cubic	
Hexagonal close packed	
	Diamond

[3 marks]

e. (i) State Le Chatelier's Principle.

(ii) Consider the reaction

$$\mathrm{2NO_2(g) \rightleftharpoons N_2O_4(g)}\ \Delta H^{\ominus} = +58.1\ \mathrm{kJ\ mol^{-1}}$$

How does an increase in temperature affect:

A. The equilibrium constant?

B. The equilibrium position? *[3 marks]*

f. Use the data below, where appropriate to answer the following questions:

	Standard Electrode Potential	$E^{\ominus}$/V
A	$\mathrm{Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)}$	+0.27
B	$\mathrm{NO_3^-(aq) + 3H^+(aq) + 2e^- \rightarrow HNO_2(aq) + H_2O(l)}$	+0.94
C	$\mathrm{Cl_2(aq) + 2e^- \rightarrow 2Cl^-(aq)}$	+1.36
D	$\mathrm{Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)}$	+1.33

	Standard Electrode Potential	$E^{\ominus}$/V
E	$S_2O_8^{2-}(aq) + 2e^- \rightarrow 2SO_4^{2-}(aq)$	+2.01

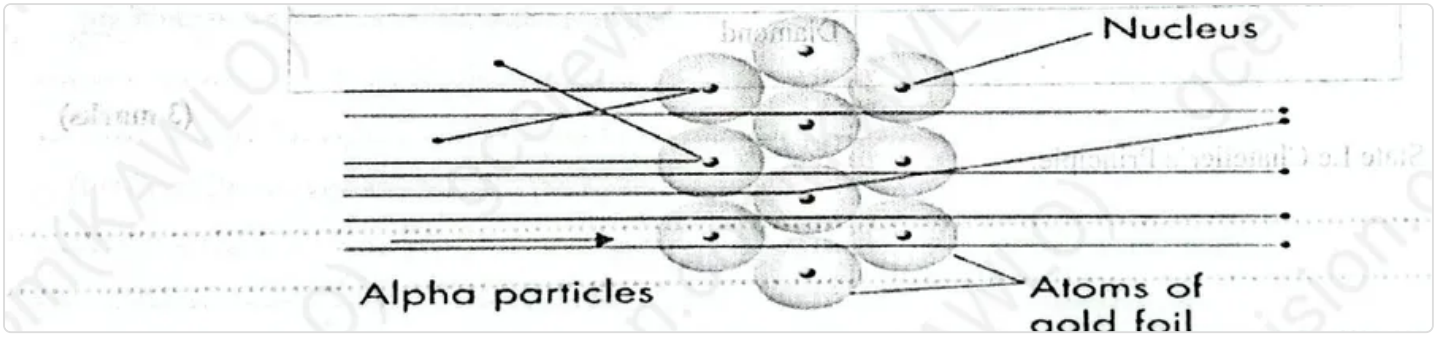
(i) From the table above, select the species which is the most powerful reducing agent.

(ii) Calculate the oxidation state of Chromium in $Cr_2O_7^{2-}$.

(iii) Write a cell diagram for the reaction when A and D are coupled and calculate the emf of the cell. *[4 marks]*

Exercise 2 (20 marks)

a. In the *"scattering of alpha particles experiment"*, the results obtained by Geiger and Marsden can be summarized by the diagram below;



(i) Give the mass number of an alpha particle?

(ii) Explain why a few of the alpha particles were deflected at large angles?

(iii) Explain why most of the alpha particles went through undeflected? *[3 marks]*

b. The half-life of tritium, a radioactive isotope of hydrogen is 12.25 years. Beginning with a 2.5 g sample of tritium, what mass will be left after 45 years? *[2 marks]*

c. The Collision Theory of Reaction Kinetics was developed from the kinetic theory of gases.

(i) State two postulates of the Collision Theory.

(ii) Hydrogen peroxide decomposes according to the equation:

$$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g).$$

State and explain one method by which the rate of the reaction can be measured

(iii) Given that the rate law for the reaction is $Rate = K[H_2O_2]^2$. What does K represent *[5 marks]*

d. Complete the table below by filling in the missing information

Bond type	Species involved	Example of compound with the type of bond
Ionic bond		
Covalent		

Bond type	Species involved	Example of compound with the type of bond
Dative covalent bond		

[3 marks]

e. Suggest an explanation for the trigonal pyramidal shape of $\mathrm{NH_3}$. *[2 marks]*

f. Two pure liquids A and B, have vapour pressures of $1.5 \times 10^4\ \mathrm{N\,m^{-2}}$ and $3.5 \times 10^4\ \mathrm{N\,m^{-2}}$ respectively at $20^\circ\mathrm{C}$. If a mixture of the two liquids is prepared by adding 2.5 moles of A and 1.5 moles of B at the same temperature, calculate:

(i) The partial vapour pressure of A and B in the mixture:

(ii) The total vapour pressure of the mixture. *[3 marks]*

g. A solution made up of $0.15\ \mathrm{mol/dm^3}$ propanoic acid ($\mathrm{K_a} = 1.35 \times 10^{-5}\ \mathrm{mol/dm^3}$) and $0.35\ \mathrm{mol/dm^3}$ sodium propanoate, constitutes a buffer solution.

(i) What is a buffer solution?

(ii) Calculate the pH of the solution. *[2 marks]*

Exercise 3 (20 marks)

a. The elements of period three of the periodic table are; sodium, magnesium, Aluminium, silicon, Phosphorus, Sulphur, chlorine and Argon.

(i) Sketch a graph to show the variation of Atomic Radius against atomic number for the elements.

(ii) Explain the shape of the graph in (i) above.

(iii) Complete the table below by either giving the structural form or the formula of an oxide.

Structural form	Formula of oxide
Ionic	
	$\mathrm{SiO_2}$
	$\mathrm{Cl_2O}$

[7 marks]

b. The complex ions of the elements of the first transition series undergo isomerism.

(i) Define the following;

A) Complex ion:

B) Isomerism:

(ii) Draw structures to show Geometric isomerism in the complex $\mathrm{Co\,(NH_3)_4(H_2O)_2}$.

Isomer 1 / Isomer 2

(iii) Using the spdf notation, write the electronic configuration of the following ions;

$_{21}Sc^{3+}$:

$_{25}Mn^{2+}$:

(iv) Which of the ions in (iii) above is not coloured? Explain. *[8 marks]*

c. The Contact Process for the production of sulphuric acid involves three stages; the production of sulphur dioxide, sulphur trioxide and sulphuric acid. Write balanced equations for the processes involved in each stage.

Stage 1:

Stage 2:

Stage 3: *[3 marks]*

d. Nitrogen exhibits oxidation states of -3 to +5 in its compounds. Complete the table below;

Oxidation state of nitrogen	Formula of compound
-3	
	NH_2OH
+2	
+5	

[2 marks]

Exercise 4 (20 marks)

a. (i) State and explain the trend in the hydration energies of the group II sulphates down the group

(ii) Write an equation for the reaction of Barium with water

(iii) Would the reaction in a (ii) occur more vigorously or less vigorously than the reaction of Calcium with water? Explain

(iv) At a given high temperature, which of the two carbonates, Barium carbonate or Calcium carbonate, would decompose more easily? *[6 marks]*

b. The Group IV (14) elements: Carbon (C), Silicon (Si), Germanium (Ge), Tin (Sn) and Lead (Pb), have melting points $3730^\circ C$, $1410^\circ C$, $937^\circ C$, $232^\circ C$ and $327^\circ C$ respectively.

(i) Give the trend in the structure of the elements

(ii) Explain the variation in the melting points

(iii) Account for the fact that carbon tetrachloride is a insoluble in water. *[7 marks]*

c. (i) Complete the table below

halogen	physical state	colour
Fluorine		
Chlorine		
Bromine		
Iodine		

(ii) Write a balanced equation to show what happens when chlorine gas is bubbled into an aqueous solution of potassium bromide (KBr)

(iii) What are the observable changes?

(iv) An aqueous solution containing an unknown halide ion, was acidified with nitric acid; aqueous silver nitrate was then added. A cream precipitate was obtained which dissolves in concentrated aqueous ammonia. Identify the halide ion *[7 marks]*

Exercise 5 (20 marks)

a. The mass spectrum of a compound A has parent ion peak at m/e = 168. Elemental analysis shows that it contains 42.9% Carbon; 2.4% hydrogen and 16.7% nitrogen by mass. The remainder is oxygen

(i) Calculate the empirical formula of the compound A

(ii) Determine the molecular formula of the compound A

(iii) Briefly describe how you can test for the presence of Nitrogen in compound A. *[7 marks]*

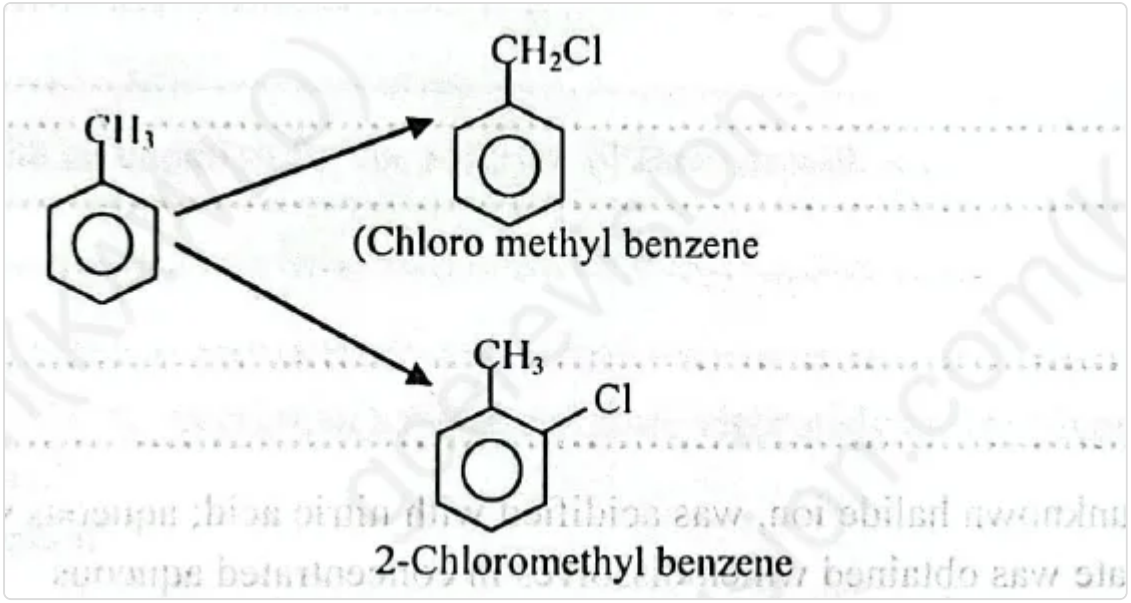
b. Write down the systematic name of each compound below.

(i) $(CH_3)_3COH$.

(ii) $CH_3CH(Br)CH_2CH_3$.

(iii) $CH_3(CH_2)_2CHCH_2$. *[3 marks]*

c. Under different reaction conditions, methylbenzene reacts with chlorine by different reaction mechanisms. One product of each reaction is shown below



(The scheme shows methylbenzene giving, in the upper reaction, the product with CH_2Cl on the ring (printed name *"Chloro methyl benzene"*) and, in the lower reaction, the product with CH_3 and Cl on adjacent ring carbons (printed name *"2-Chloromethyl benzene"*).)

(i) For the reaction leading to the formation of Chloromethyl benzene;

(A) Give the reaction condition

(B) Name the type of reaction mechanism involved

(ii) For the reaction leading to the formation of 2-chloromethyl benzene.

A: Give the reaction condition

B: Name the type of reaction mechanism involved *[4 marks]*

d. Aldehydes and ketones undergo nucleophilic addition reactions with HCN.

(i) What do you understand by a nucleophilic addition reaction?

(ii) Using ethanal, (CH_3CHO) write the mechanism for the reaction with HCN.

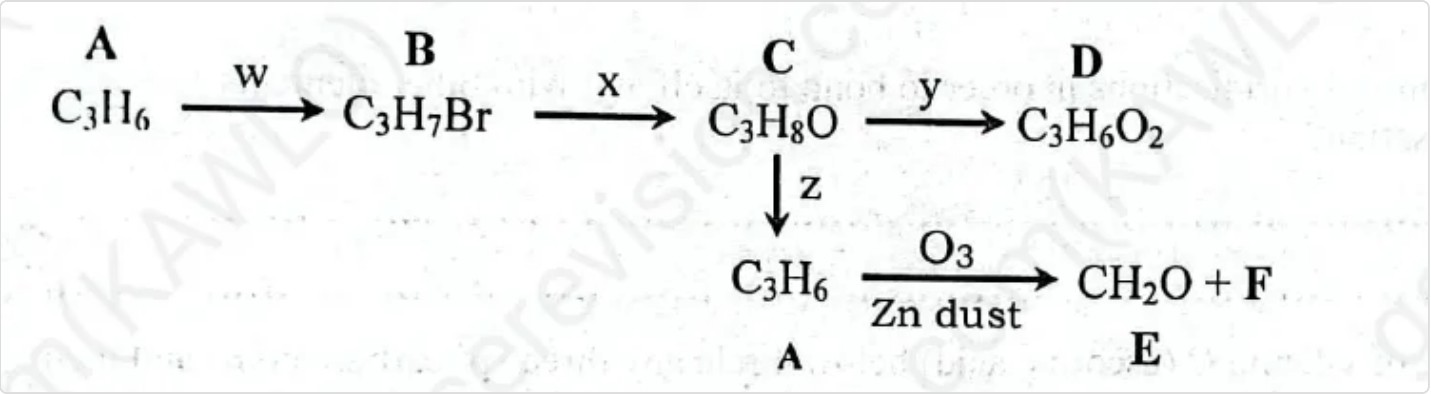
(iii) Name the functional group present in aldehydes and ketones *[4 marks]*

e. (i) What is a functional group?

(ii) Name the functional group present in aldehyde and ketones. *[2 marks]*

Exercise 6 (20 marks)

a. Study the reaction scheme below and answer the questions that follow.



(The scheme: A (C_3H_6) reacts by step w to give B (C_3H_7Br); B reacts by step x to give C (C_3H_8O); C reacts by step y to give D ($C_3H_6O_2$); C also reacts by step z to give C_3H_6 (A); A with O_3 then Zn dust gives CH_2O (E) + F.)

(i) Write down the structural formulae of compounds A, B, C, D and F.

A: B: C: D: F:

(ii) Suggest the reagent(s) and reaction condition(s) for the reactions w, x, y and z.

	Reagent(s)	**Reaction condition(s)**
w		
x		
y		
z		

[9 marks]

b. How can you distinguish by a chemical test the following pairs of compounds?

(i) $CH_3CH_2COCH_2CH_3$ and $CH_3COCH_2CH_2CH_3$

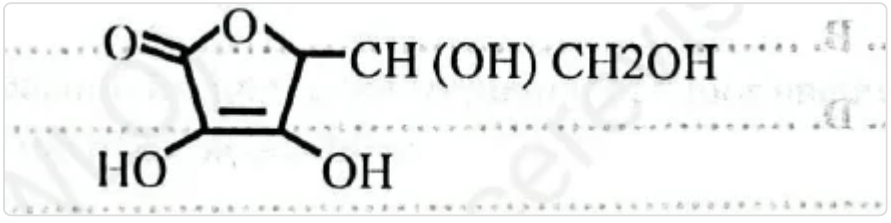
(ii) C_2H_5COOH and C_6H_5-OH

(iii) $CH_3CH_2NH_2$ and $(CH_3)_3N$ *[6 marks]*

c. Carbon undergoes sp^3, sp^2 and sp hybridisations in order to bond to itself and with other elements.

(i) What is hybridisation?

(ii) On the structure of vitamin C (ascorbic acid) below circle any three sp^3 carbon atoms and then place an asterisk on all the sp^2 hybridised carbon atoms.



(Structure as printed: a five-membered ring containing a ring oxygen, with a C=O (lactone) carbon, a C=C double bond bearing two OH groups, and a ring carbon carrying the side chain $CH(OH)CH_2OH$.) *[3 marks]*

d. Suggest the spectroscopic methods that could be used to identify the following:

(i) Positions of hydrogen atom.

(ii) Presence of double bonds. *[2 marks]*

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